Learning invidivdual intrinsic rewards (LIIR) [Du et al. 2019]

- learn intrinsic reward per agent
  - MLP (η) with 2 hidden layers [Zheng, Oh, and Singh 2018]
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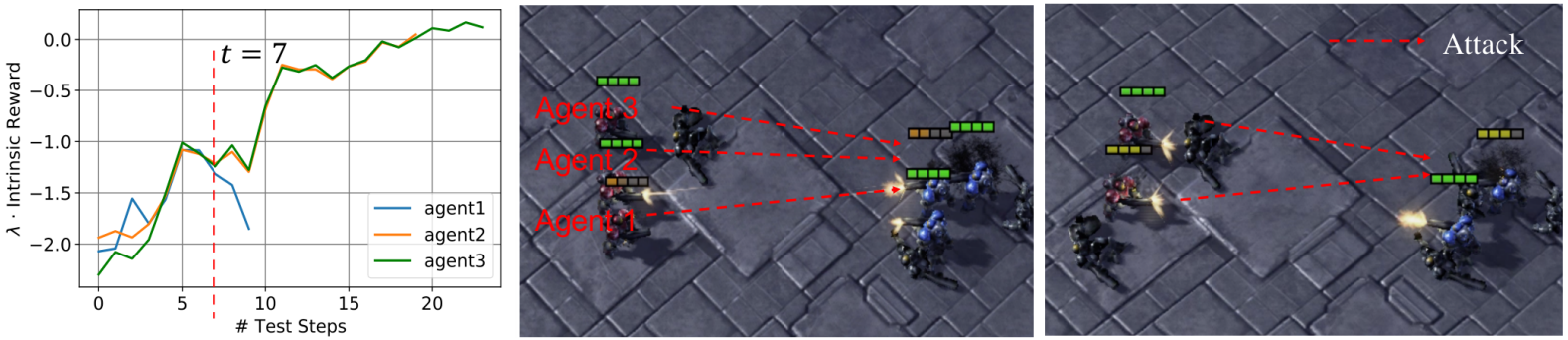
- update θ with regards to proxy reward
- update η with regards to extrinsic rewards via meta-gradient

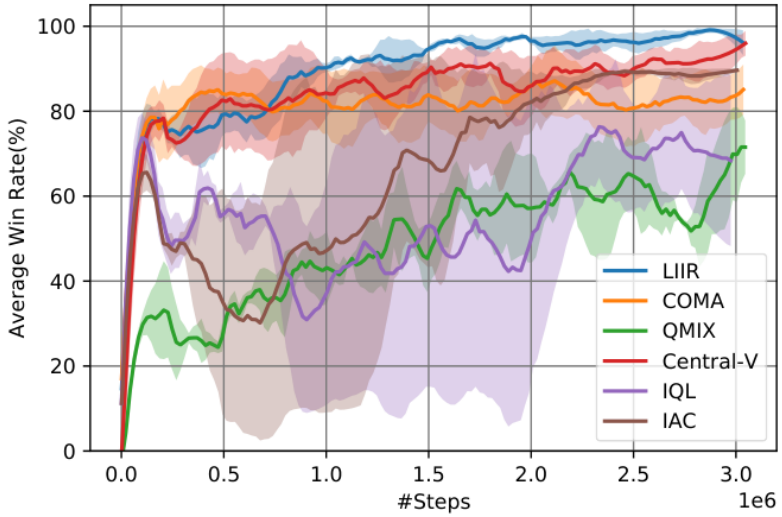


(a) Intrinsic reward (b) $t = 6$ (c) $t = 9$

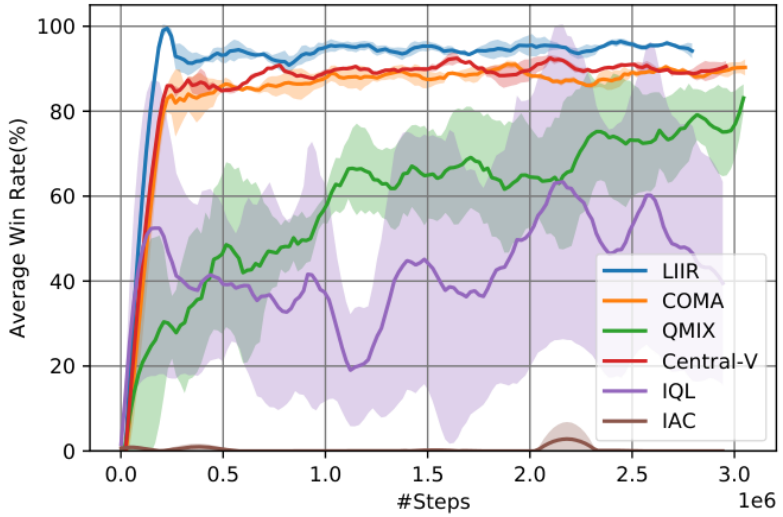
Figure 4: An example of the intrinsic reward curves and auxiliary snapshots on 3M.

(a) Intrinsic reward (b) $t = 27$ (c) $t = 32$

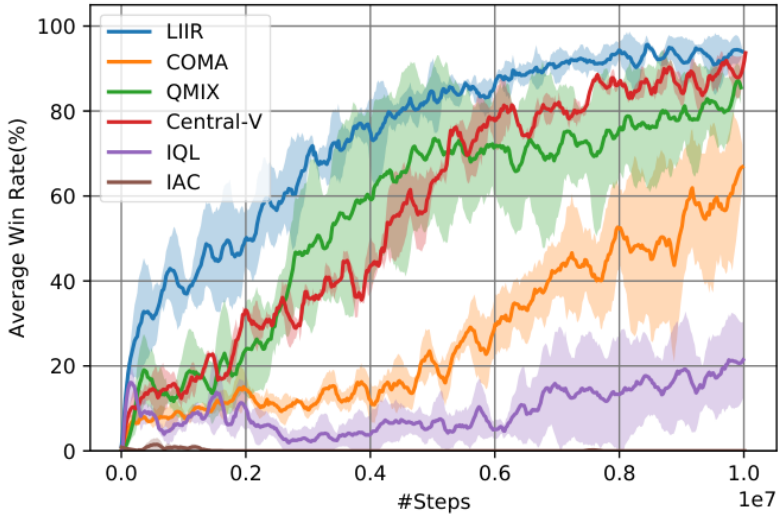
Figure 5: An example of the intrinsic reward curves and auxiliary snapshots on 2S3Z.



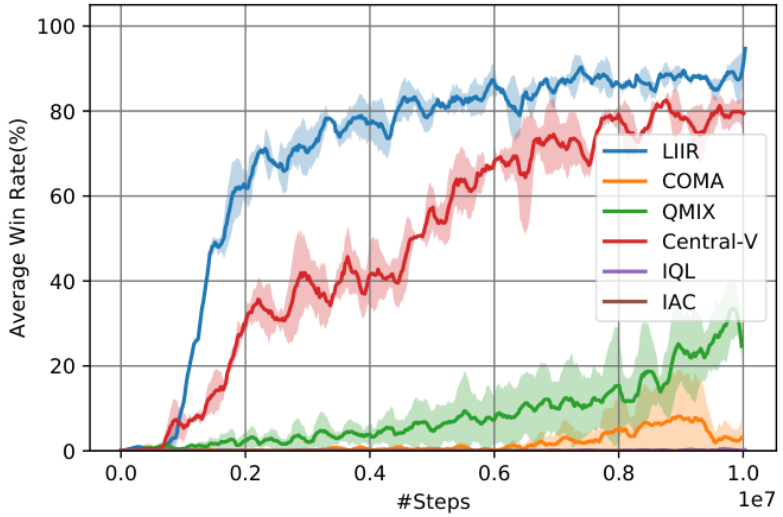
(a) 3M



(b) 8M



(c) 2S3Z



(d) 3S5Z

Figure 3: Test winning rates vs. training steps of various methods on all the scenarios.